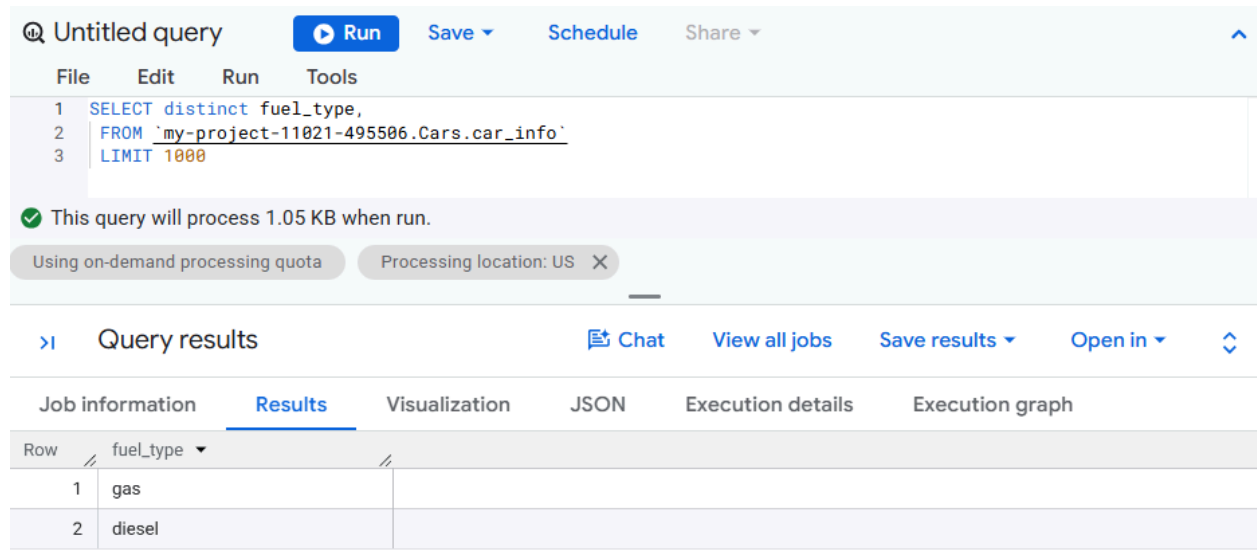


CARS DATASET : clean data using SQL

Ensuring data integrity and cleanliness to prevent inaccurate vehicle reporting , mitigating the risk of flawed inventory investment and misinformed business decisions.

TASK 1: confirm that the `fuel_type` column doesn't have any unexpected values.



The screenshot shows a SQL query editor interface. At the top, there's a search bar with 'Untitled query' and buttons for 'Run', 'Save', 'Schedule', and 'Share'. Below the search bar are tabs for 'File', 'Edit', 'Run', and 'Tools'. The SQL query is as follows:

```
1 SELECT distinct fuel_type,  
2 FROM `my-project-11021-495506.Cars.car_info`  
3 LIMIT 1000
```

Below the query, a green checkmark icon indicates 'This query will process 1.05 KB when run.' There are also buttons for 'Using on-demand processing quota' and 'Processing location: US'.

Below the query editor, there's a section for 'Query results' with buttons for 'Chat', 'View all jobs', 'Save results', and 'Open in'. Below this, there are tabs for 'Job information', 'Results', 'Visualization', 'JSON', 'Execution details', and 'Execution graph'. The 'Results' tab is selected, showing a table with two columns: 'Row' and 'fuel_type'. The table contains two rows of data:

Row	fuel_type
1	gas
2	diesel

Key insights :

- Unique value found : The `fuel_type` column contains only two distinct categories: "gas" and "diesel"
- Data Quality Check: There are no unexpected values, typos, or extra spaces visible in the fuel types at this stage, indicating that this particular column is relatively clean.

TASK 2 : check the minimum and maximum lengths in the dataset.

Untitled query X Job history X *Untitled query X X

Untitled query Run Save Schedule Share

File Edit Run Tools

```

1 SELECT
2 MIN(length) AS min_lenght,
3 MAX(length) AS max_lenght
4 FROM `my-project-11021-495506`.Cars.car_info

```

✓ Query completed

Using on-demand processing quota Processing location: US X

> Query results Chat

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	min_lenght	max_lenght			
1	141.1	208.1			

Key Insight:

a. Length Range Found: The length column ranges from a minimum value of 141.1 to a maximum value of 208.1.

B. Data Verification: Checking the min and max values helps confirm that the numerical data falls within a realistic and expected scale without any extreme or erroneous outliers.

TASK 3 : check your data for null or missing values. These values might appear as a blank cell or the word NULL in BigQuery.

Untitled query Run Save Schedule Share

File Edit Run Tools

```

1 SELECT *
2 FROM `my-project-11021-495506`.Cars.car_info
3 WHERE
4 num_of_doors IS NULL

```

✓ This query will process 28.71 KB when run.

Using on-demand processing quota Processing location: US X

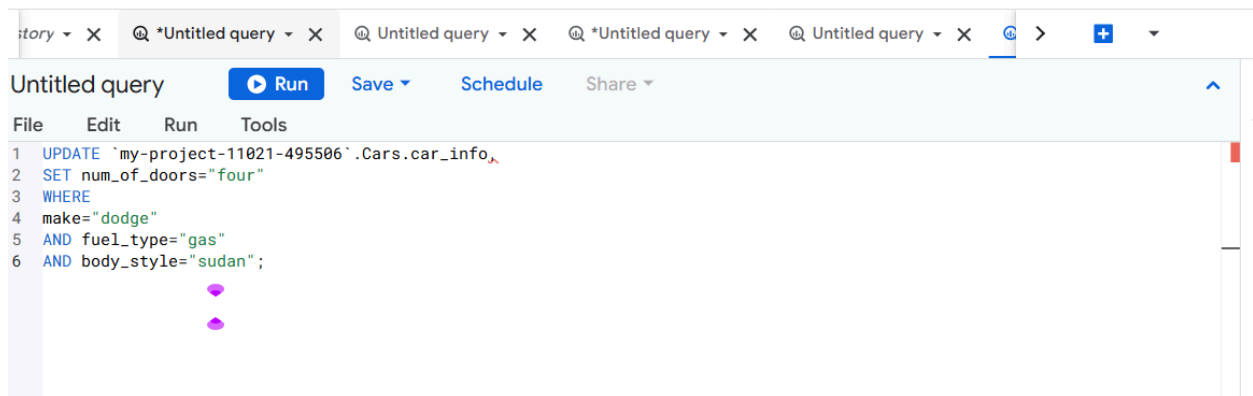
> Query results Chat View all jobs Save results Open in

Job information	Results	Visualization	JSON	Execution details	Execution graph	
Row	make	fuel_type	num_of_doors	body_style	drive_wheels	engine
1	dodge	gas	null	sedan	fwd	fr
2	mazda	diesel	null	sedan	fwd	fr

Key Insight:

- Missing Values Found: Identified records where the num_of_doors column has null values.
- Action Required: These missing values need to be cleaned or updated before analysis.

TASK 4 : Order to fill in these missing values, check with the sales manager, who states that all Dodge gas sedans and all Mazda diesel sedans sold had four doors .

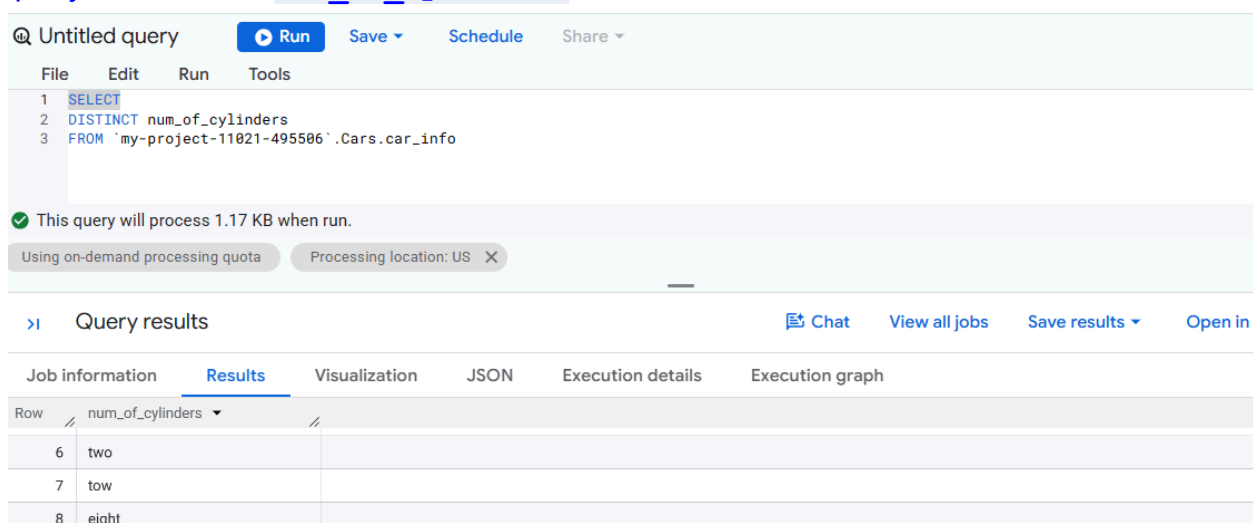


```
1 UPDATE `my-project-11021-495506`.Cars.car_info
2 SET num_of_doors="four"
3 WHERE
4 make="dodge"
5 AND fuel_type="gas"
6 AND body_style="sudan";
```

Key Insight:

- Data Updated: Successfully filled in the missing door counts with "four" for the specified Dodge models.
- Data Cleaning Complete: Resolved missing value issues for targeted records using business rules.

TASK 5 : check for other potential errors to check what values exist in a column. Run a query to check the num_of_cylinders column:



```
1 SELECT
2 DISTINCT num_of_cylinders
3 FROM `my-project-11021-495506`.Cars.car_info
```

✓ This query will process 1.17 KB when run.
Using on-demand processing quota Processing location: US

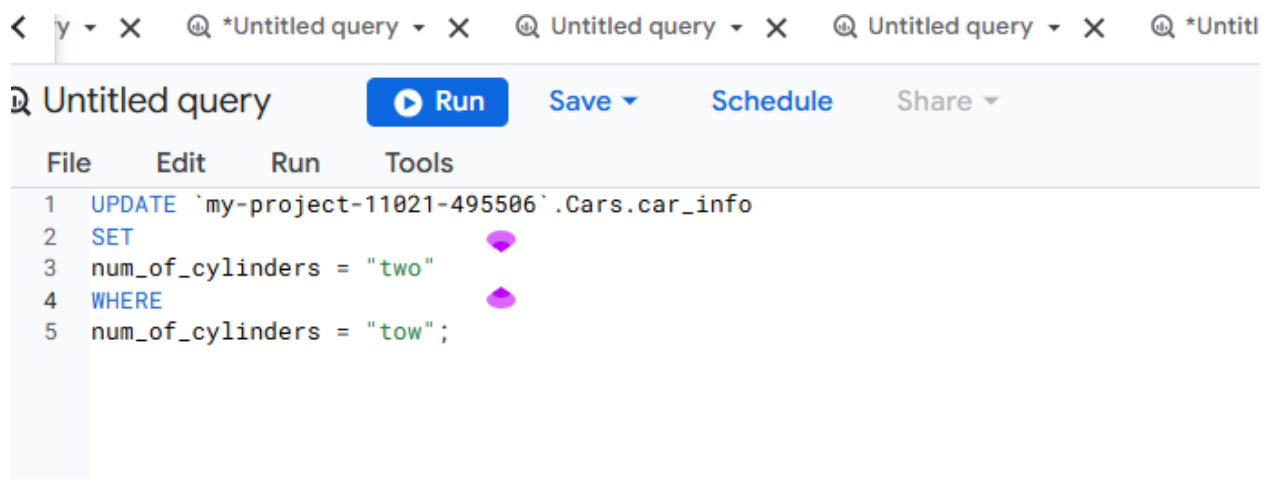
Query results Chat View all jobs Save results Open in

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	num_of_cylinders				
6	two				
7	tow				
8	eight				

Key Insight:

- a. Data Errors Found: The num_of_cylinders column contains typos and inconsistent spelling entries (such as "two" and "tow").
- b. Data Cleaning Required: These spelling mistakes need to be corrected to standardize the cylinder categories before analysis.

TASK 6 : To correct the misspelling for all rows



The screenshot shows a SQL query editor interface. At the top, there are several tabs labeled "Untitled query". Below the tabs, there is a menu bar with "File", "Edit", "Run", and "Tools". The "Run" button is highlighted. The main area contains a SQL query with line numbers 1 through 5. The query is as follows:

```
1 UPDATE `my-project-11021-495506`.Cars.car_info
2 SET
3 num_of_cylinders = "two"
4 WHERE
5 num_of_cylinders = "tow";
```

Key Insight: Successfully standardized the num_of_cylinders column by correcting the spelling error ("tow" to "two"). This ensures data consistency and prevents aggregation errors in future queries and analysis.

TASK 7: The **compression_ratio** column values should range from 7 to 23. Just like when you checked the length values, use **MIN** and **MAX** to check if that's correct, we notice **this returns a maximum of 70**. This is an error because the maximum value in this column should be 23, not 70. So the 70 is most likely a 7.0. Run the query without the row with 70 to make sure that the rest of the values fall within the expected range of 7 to 23.

Untitled query Run Save Schedule Share

File Edit Run Tools

```

1 SELECT
2 MIN(compression_ratio) AS min_compression_ratio,
3 MAX(compression_ratio) AS max_compression_ratio
4 FROM `my-project-11021-495506`.Cars.car_info
5 WHERE
6 compression_ratio <> 70;

```

✓ Query completed

Using on-demand processing quota Processing location: US X

> Query results Chat

Job information	Results	Visualization	JSON	Execution details	Execution g
Row	min_compressio...	max_compressio...			
1	7.0	23.0			

Key Insights: Filtered out the outlier value (70) where compression ratio was incorrectly entered, and found that the valid compression ratio ranges from a minimum of 7.0 to a maximum of 23.0.

TASK 8: The rows should be deleted containing 70 as compression ratio.

ery X Untitled query X *Untitled query X *Untitled query X Untitled query

Untitled query Run Save Schedule Share

File Edit Run Tools

```

1 DELETE FROM `my-project-11021-495506`.Cars.car_info
2 WHERE compression_ratio = 70 ;

```

Key Insight : the corresponding rows has been deleted has run the query again to check.

TASK 9 : Check the `drive_wheels` column for inconsistencies by running a query with a `SELECT DISTINCT` statement:

Untitled query Run Save Schedule Share

File Edit Run Tools

```

1 SELECT
2 DISTINCT drive_wheels
3 FROM `my-project-11021-495506`.Cars.car_info

```

Query completed

Using on-demand processing quota Processing location: US

Query results Chat View all jobs

Job information Results Visualization JSON Execution details Execution graph

Row	drive_wheels
2	fwd
3	4wd
4	4wd

Key Insight : It appears that 4wd appears twice in results. However, because you used a **SELECT DISTINCT** statement to return unique values, this probably means there's an extra space in one of the 4wd entries that makes it different from the other 4wd.

TASK 10 : check if this is the case, you can use a **LENGTH** statement to determine the length of how long each of these string variables:

Untitled query Run Save Schedule Share

File Edit Run Tools

```

1 SELECT
2 DISTINCT drive_wheels,
3 LENGTH (drive_wheels) AS string_lenght
4 FROM `my-project-11021-495506`.Cars.car_info

```

Query completed

Using on-demand processing quota Processing location: US

Query results Chat View al

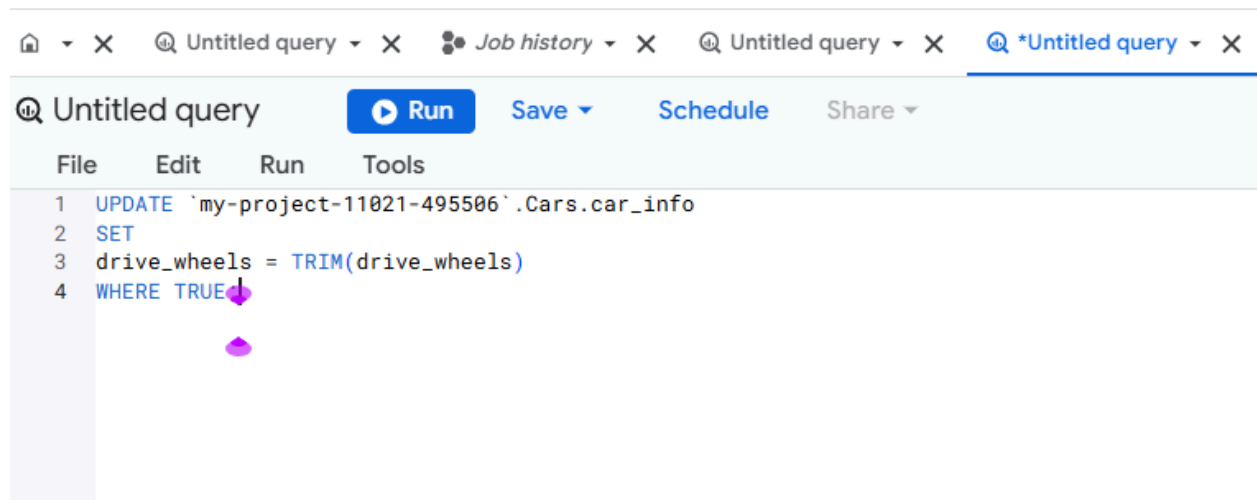
Job information Results Visualization JSON Execution details Execution graph

Row	drive_wheels	string_lenght
2	fwd	3
3	4wd	4

Key Insights:

- a. Evaluated string lengths using the LENGTH() function alongside DISTINCT on the drive_wheels column to analyze the unique text values and their exact character counts.
- b. Confirmed that the distinct categories are fwd (with a string length of 3 characters) and 4wd (with a string length of 4 characters).
- c. Observed that we should use the TRIM function to remove all extra spaces in the drive_wheels column.

TASK 11 : Use the TRIM function to remove all extra spaces in the drive_wheels column.



The screenshot shows a SQL query editor interface. At the top, there are several tabs: 'Untitled query', 'Job history', 'Untitled query', and '*Untitled query'. Below the tabs is a toolbar with buttons for 'Run', 'Save', 'Schedule', and 'Share'. The main area displays a SQL query with line numbers 1 through 4. The query is an UPDATE statement that sets the drive_wheels column to the TRIM of its current value for all rows where the condition is TRUE.

```
1 UPDATE `my-project-11021-495506`.Cars.car_info
2 SET
3 drive_wheels = TRIM(drive_wheels)
4 WHERE TRUE
```

KEY INSIGHT :

- a. The TRIM() function has eliminated extra leading and trailing spaces that caused inconsistent character counts in your previous LENGTH analysis.
- b. This ensured clean, uniform formatting across all values in the drive_wheels column.
- c. Improved Accuracy and queries can run accurately without errors caused by invisible spaces.